

AZIZ ULLAH

Senior Python Developer • Web Scraping • Data Engineering • AI API Integration

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PROFESSIONAL SUMMARY

Python Backend Developer with **5 years of experience** building reliable data pipelines, RESTful APIs, web scraping systems, and AI-assisted automation tools. Specialized in Python (Flask, FastAPI, Pandas) with strong hands-on expertise in extracting structured data from complex dynamic websites and integrating third-party AI APIs (OpenAI, Anthropic) into production applications. Experienced in designing clean architectures, handling high-volume datasets, and building resilient backends with solid logging, error handling, and performance optimization.

TECHNICAL SKILLS

Core Languages:	Python 3.x, SQL
Backend Frameworks:	FastAPI, Flask, RESTful API Design, Webhooks, JSON APIs
Web Scraping & Auto:	Selenium, BeautifulSoup, Puppeteer, Async Scraping, Proxy Rotation, User-Agent Spoofing, Rate-Limit Handling
Data Engineering:	Pandas, Data Cleaning, Schema Normalization, CSV/JSON Processing, Vectorized Operations
AI & LLM Integration:	OpenAI API, Anthropic Claude API, Structured Outputs, Function Calling, Prompt Engineering, AI Dev Tools (Cursor, Copilot)
Databases & Tools:	PostgreSQL, SQLite, MySQL, Redis (basics), Git, Docker, Linux/Bash, Postman, Pytest

PROFESSIONAL EXPERIENCE

Senior Python & AI Backend Developer — Smart Forum

Jan 2022 – Present

- Designed and managed scalable web scraping systems to extract structured data from JavaScript-heavy, dynamic websites across multiple target domains.
- Integrated AI models (OpenAI API and Claude API) into data extraction pipelines to parse unstructured web content into validated JSON schemas, significantly improving parsing accuracy for non-standard page layouts.
- Developed high-performance RESTful APIs using FastAPI and Flask to deliver structured data and service integrations to downstream web applications.
- Built end-to-end data processing pipelines using Pandas for cleaning, deduplicating, transforming, and validating large datasets before database storage.
- Engineered resilient scraping architectures with automatic retry mechanisms, exponential backoff, proxy management, and comprehensive logging to maximize uptime and bypass rate limiting.
- Leveraged prompt engineering techniques and AI development tools (Cursor, Copilot) to speed up routine script generation, unit test creation, and codebase refactoring.
- Automated core client data workflows, reducing manual operational effort by up to 90% while enhancing overall system execution speed and data reliability.

Python Developer (Backend & Automation) — Software Development Consultant

Jan 2020 – Dec 2021

- Built custom Python scripts for data collection, automated report generation, and website data extraction for various business clients.
- Developed REST API endpoints using Flask to support client data ingestion and database CRUD operations.
- Processed, normalized, and transformed messy data exports (CSV, JSON, XML) using Python data structures and Pandas to ensure clean inputs for analysis.
- Designed database schemas in SQLite and MySQL to reliably store scraped and user-generated data.
- Implemented error handling, input validation, and scheduled background tasks to ensure automated jobs ran reliably without manual supervision.

KEY PROJECTS

AI-Enhanced Modular Web Scraping Framework

Technologies: Python, Selenium, BeautifulSoup, OpenAI API, Pandas, FastAPI

- Built a modular scraping engine supporting authentication, paginated navigation, and structured JSON parsing via OpenAI function calling. Reduced new scraper setup time by 60%.

FastAPI Data Delivery REST API

Technologies: Python, FastAPI, Pydantic, PostgreSQL, Swagger / OpenAPI

- Designed and deployed lightweight REST API services with Pydantic request validation and async handlers for high concurrency and low-latency JSON delivery.

Automated Data Processing & ETL Pipeline

Technologies: Python, Pandas, NumPy, JSON/CSV

- Developed automated Python ETL scripts using vectorized Pandas operations for cleaning and merging multi-source datasets, replacing manual spreadsheet operations.

EDUCATION & LANGUAGES

Bachelor of Science in Information Technology (BS IT) — University of Malakand, Pakistan (CGPA: 3.12 / 4.0) 2015 – 2019

Languages: English (Professional Working Proficiency), Urdu (Native), Pashto (Native)